Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Zsolt Kira	**Email**	zkira@gatech.edu
Institute Name	Georgia Institute of Technology			
Address	756 W Peachtree St NW, Atlanta, GA 30308			

2. Technology Overview

Title of Proposed Technology	***Lifelong Personalized Home Robots that Continually Remember, Reason, and Learn***			
Korea's 12 National Strategic Technology Fields	**Semiconductor/Display (), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (X), Next-gen Communication ( ), Advanced Robotics & Manufacturing (v), Quantum ( )**			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Mechanical ・ Materials			
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	***Robot Control and Intelligence Technologies***	*Robotics/ Automation Machinery*	Robot Vision and Production Automation Technologies
Concept of Proposed Technology	The proposed technology aims to develop algorithms, deep learning architectures, and training methods to enable lifelong home robots to continually remember past experiences and tasks, reason about and retrieve such memories during new requests, and distill them into new skills over time. Recently, a flurry of research has produced effective robot policies that can perform a number of tasks in a home, ranging from cleaning up the home to putting away the groceries [1-2]. This includes the development of Vision-Language-Action models (including in our group [2-3]), where Multimodal Large			

	Language Models (such as open-source models similar to ChatGPT) are further trained (fine-tuned) to generate robot actions rather than language. Such models are typically trained with human demonstrations of tasks, and then optionally further trained through experiences gathered by the robot itself through reinforcement learning (RL) [2]. While effective, the resulting models lack a number of important capabilities that humans have and are crucial for deployment in environments such as the home. Specifically, such models are pre-trained offline and then deployed, lacking the ability to personalize to the user (i.e. learn and understand their preferences, desired tasks, etc.), remember past experiences, and most importantly *learn* from their experiences to improve existing skills learn new ones back into the model. This precludes robots that can become more effective over time as they interact in a particular home and with a particular user. In this project, **we propose to develop novel mechanisms for robots to:** **1) Remember their experiences** - Synthesize experiences and human demonstrations or feedback into structured, compressed memory representations, 2) **Reason about their experiences** – Given new queries or tasks, reason about past experiences that are relevant and condition their policies on such memories, and 3) **Learn new skills** – Distill past experiences into updates to the underlying model itself such that new frequently-requested behaviors are baked directly into the model, rather than requiring retrieval or reasoning mechanisms.
Rationale for Joint R&D Project	As robots become more capable, they are poised to enter homes for a range of uses including elderly care, personal assistants that can do chores, or entertainment.. However, for robots to be truly useful they must learn new skills through experience as they encounter new requests from users. Indeed, the range of tasks that users can request, including with personalized preferences (e.g. where particular objects should go in a home, how to clean up the home, where to stay away from, etc.), is extremely large. While current works attempt to simply scale as many human demonstrations as possible (including those collected across many different embodiments and universities/industries), such scale is extremely difficult to achieve given the embodiment differences between different robots, the huge variety of environments and tasks, and the existence of human preferences in carrying out the same task. In order to enable this research, industrial partners are needed to provide robot hardware and infrastructure, as well as real-world environments, in order to gather data and test the robot's abilities to learn from experience. Indeed, the Korean SMEs will be well-positioned to define the use-cases and scenarios that will serve as the test domain, representing real-world tasks that humans would desire of such home robots that have high potential to help society and for commercialization. The Korean SME can provide such infrastructure, while we develop fundamental algorithms to drive them as the university partner.
Key Words	1. *Home robotics* 2. *Cognitive architectures for robotics* *3.* Lifelong Robotics

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]		Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives		The objectives of the research are to develop novel learning architectures, algorithms, and training pipelines to enable robots to remember new experiences, reason over them, and learn new tasks after deployment into a home environment. This includes: - Development of novel memory architectures that can be incorporated into Vision-Language-Action models including learnable compressed latent memories that provide a structured compression of long raw observations (i.e. image/video streams, proprioception, etc.). - Development of novel retrieval and reasoning mechanisms to obtain the appropriate compressed memories when being requested to carry out a new task or improve performance on an existing task. - Development of novel lifelong/continual learning algorithms that can use these retrieved memories as training data in order to update the model directly such that it can natively carry out these tasks.			
R&D Target Specifications		The criteria to measure success in the proposed research are included below. We will leverage our prior work benchmarking memory-based tasks for home robots [4]. - <u>Memory Compression:</u> Compression ratio of the compressed memory representations compared to the original raw observations. - <u>Retrieval Performance:</u> The ability of the method to retrieve appropriate memories given a requested user task (e.g. to fetch a mug, clean up the house, etc.) will also be measured. It is expected that there will be a natural trade-off between compression (which results in some loss of information) and retrieval accuracy. - *<u>Continual/Lifelong Learning:</u>* We will measure the ability of the robot to improve on existing tasks over time (based on its past experiences) as well as learn new tasks. The metrics includes success rates on existing and new tasks, as well as a measurement of *catastrophic forgetting,* i.e. the ability of the model to not destroy its abilities on old tasks while updating existing ones or learning new ones.			
R&D Plan		Below are the milestones for the project: - **Milestone 1 - Year 1** - Development and implementation of memory methods for compression of long-form robot experiences over time, as well as simple method to utilize these memories by robot policies. In collaboration with the Korean partners, initial hardware and software infrastructure to be able to run simple mobile manipulation tasks in home environments. - Simulation and real-world based development and testing, with a final real-world demonstration at the end of the year.			

		Milestone 2 - Year 2: - Development of novel methods for leveraging compressed memory representations for improved task execution, with improvements in task success rates over time. - Continued integration into the hardware platforms for testing, with a few real-world useful tasks that can leverage memory, such as retrieval of lost objects (e.g. “I can’t find my keys – can you remember where they are and fetch them for me?”. **Milestone 3 - Year 3:** - Development of methods to continuously learn from past experiences and improve existing tasks over time as well as learn new tasks as demonstrated by humans. - Integration of methods on real robots to demonstrate the robot’s capability of learning new tasks specified or demonstrated by the user. - Assessment of commercialization potential across a range of domains including home robotics, elderly care, etc.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Our lab will be responsible for: - Research and Development of novel methods for memory creation, reasoning, and continual learning. - Implementation and delivery of code, data, and training infrastructure to enable simulation-based training and real-world finetuning of robot policies, reading for deployment (and finetuning) on real robots. - Experimentation and evaluation in simulation and real world to demonstrate the efficacy of the system. - Dissemination of papers describing the technical methodology and experimental results both to stakeholders and the community.
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for: - Design and/or selection of the robot hardware platforms (including mobile manipulator, sensors, etc.) with advisement by the research institute. - Identification of real-world useful tasks that fulfill real-world needs and have strong potential for commercialization. - Integration of robot models into the real-robot sensing and control infrastructure in order to demonstrate the resulting models on real-world tasks.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The expected outcomes of this project includes: - Research and development results including algorithms, data, code, and models reproducing our developed methods and findings. The resulting robot policy models will be deployable on both simulated and real-world robots. - Robot Infrastructure including integrated sensors, mobile base, and manipulators with code supporting the deployment of the learned models and algorithms in real-world homes. - Products such as publications describing the developed methods and experimental results. - Novel scientific developments and findings that would push a number of scientific fields forward, including computer vision, machine learning, and robotics.

6. References

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(2) A. Szot, B. Mazoure, O. Attia, A. Timofeev, H. Agrawal, D. Hjelm, Z. Gan, Z. Kira, and A.T. Toshev, "From Multimodal LLMs to Generalist Embodied Agents: Methods and Lessons", Conference on Computer Vision and Pattern Recognition, 2025.

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(4) K. Yadav, Y. Ali, G. Gupta, Y. Gal, Z. Kira, "FindingDory: A Benchmark to Evaluate Memory in Embodied Agents", NeurIPS Workshop on SPACE in Vision, Language, and Embodied AI (SpaVLE), 2025.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization